VIVEK HARI

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PROFILE SUMMARY

I have 5+ years of professional experience in software development, extended reality development, engineering simulation and workflow automation. Owned C#, C++ and Python based engineering tools, data-processing pipelines and automated workflows. Experienced in applying cross-disciplinary engineering knowledge to deliver robust, measurable and high impact software solutions.

PROFESSIONAL EXPERIENCE

Master Thesis and Working Student, Solution Architect 10/2024 - 11/2025
BMW Group Garching bei München, Germany

- Developed cross-device Unity applications for Apple Vision Pro and Meta Quest 3, enabling mixed-reality vehicle interior reviews with physical prototypes.
- Automated Blender-to-Unity XR simulation workflow for vehicle interior components using C# and Python, covering scene setup, materials, animations and interactions that reduced the setup time by 98%.
- Built Python-based Blender add-on to standardize and streamline geometry preparation and naming for downstream designer-friendly XR integration *(Incl. the conversion of shaders).*
- Implemented multi-user collaboration and real-time tracking that integrates markers and Arduino-based sensors for synchronized MR evaluations.

Working Student , ADAS/AD Function Development Engineer 06/2023 - 05/2024
CARIAD SE Ingolstadt, Germany

- Implemented the Constant Yaw Rate and Acceleration (CYRA) motion model in MATLAB, contributing to the accuracy and reliability of autonomous driving simulations.
- Reworked C++ code to calculate metrics and visualize criticality analysis methods, enhancing performance and reusability.
- Conducted thorough research on laws and regulations relevant to autonomous driving systems, ensuring compliance with legal and regulatory standards.
- Described and documented multiple driving scenarios using the Pegasus methodology, improving the test framework for software components.

Senior Structural Simulation and Automation Engineer (R&D) 08/2022 - 03/2023
Junior Structural Simulation and Automation Engineer (R&D) 09/2019 - 07/2022
MRF Tyres Chennai, India

- Led the development, testing and deployment of validated simulations in a no-code platform for the product group build using C++, Python, TCL/TK, NodeJS and PostgreSQL.
- Developed and validated simulation methodologies for evaluating static and dynamic tire characteristics (radial and cross-ply tires).
- Conducted studies to improve DFQ for cross ply tires by analysing tire building process design parameters.
- Correlated design parameters and track testing feedback to improve tire performance with testing teams.

EDUCATION

Master of Engineering, Mechatronic and Cyber-Physical Systems, Grade - 1.8 03/2023 - 01/2026
Technische Hochschule Deggendorf Cham, Germany
Thesis: Workflow Development and Automation for Prototyping Vehicle Interiors using Extended Reality

Bachelor of Technology (Honours), Mechanical Engineering, Grade - 2.0 08/2015 - 07/2019
TKM College of Engineering Kerala, India
Thesis: Design and Structural Analysis of a Below Knee CFRP Prostheses

SKILLS

Professional: Goal oriented, Effective team player, Motivated, Self-driven, Independent, Analytical & conceptual, Problem solving, Error analysis, Quick comprehension, Technical writing, Code review

Technical: CAE - Abaqus, Altair Hyperworks, ANSYS, Autodesk Fusion360, MSC Patran | CAD - SOLIDWORKS, CATIA V5, AutoCAD | Programming Languages – C#, C++, Python, VBA, MATLAB, Bash, Swift, TCL/TK | Robotics – ROS1 Noetic, Gazebo, Moveit, Movebase, AMCL, SLAM | Computer Vision – OpenCV, PyTorch | System Modelling – SIMULINK | Machine Learning - PyCaret, Pandas, NumPy, Matplotlib, Seaborn | IDEs - PyCharm, Spyder, Jupyter Notebook, Visual Studio Code | Operating Systems – Windows, Linux (Ubuntu) | Documentation – Confluence, MS PowerPoint, MS Excel, MS Word | Game Engine – Unity, Unreal Engine, Godot | DevOps - Jira, Git, OOPs, CMake, Azure, Agile, CI/CD, Sprint Planning (Scrum), User Story Writing

Language: English – C1 (Fluent); German – A2 (Beginner, Pursuing B1); Malayalam – Mother Tongue, Tamil – Only Spoken

CERTIFICATIONS

ROS1x: Hello (Real) World with ROS (Delft University of Technology) | **The C++20 Masterclass : From Fundamentals to Advanced** (Udemy) | **Automated and Connected Driving Challenges** (RWTH Aachen University) | **Mastering Programming with MATLAB**, (Vanderbilt University) | **Machine Learning** (University of Stanford) | **Python for Everybody Specialization** (University of Michigan) | **A Hands-on Introduction to Engineering Simulation** (Cornell University)

PERSONAL INTERESTS

Enthusiastic about continuous learning and applying new skills across disciplines. Passionate about Formula 1, cinematic storytelling, and interactive digital experiences through computer gaming.

Regensburg, 27/01/2026

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